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Ingestion Team
AI-Led Development
Environment

Claude Code + GitHub Copilot + New Relic MCP

🚀 10× faster feature delivery

🧠 Onboard in hours, not weeks

🔍 30-min incident diagnosis

⚡ 5 agents working in parallel

📦 Knowledge that compounds

The Challenge — Four Pains That Slow Every Team Down

🕐 **New Engineer Onboarding**

2+ WEEKS

to reach full productivity

🔍 **Incident Root Cause**

2–3 HOURS

hunting logs manually at 2am

💥 **Blast Radius Unknown**

5-DAY

incident from a '1-day fix'

🗃️ **Architecture in Heads**

KNOWLEDGE

lost every time someone leaves

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Two AI tools. One team environment. Knowledge that compounds every session.

GitHub Copilot — Line-level AI

- ✅ Autocomplete lines & functions in IDE
- ✅ Inline chat: explain / refactor this
- ✅ Generate tests & boilerplate fast
- ✅ Works where engineers already work
- ❌ No memory across sessions
- ❌ Can't query NR, Jira, Confluence
- ❌ No blast-radius or impact analysis
- ❌ No multi-agent parallel execution

+

together

Claude Code — Team environment AI

- ✅ Memory: full context loads at session start
- ✅ NR MCP: queries logs, metrics, alerts live
- ✅ Blast radius: impact analysis before coding
- ✅ /avengers: 5 parallel AI agents on one task
- ✅ /golden + /replay: patterns that compound
- ✅ 20 skills auto-trigger on file type / topic
- ✅ Jira + Confluence: auto-updated every session
- ✅ Budget dial: daily spend awareness built-in

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The Full AI-Led Development Lifecycle — From Idea to Knowledge

01	02	03	04	05	06	07	08	09	10
Requirement	**Spec & Plan**	**Architect**	**Build**	**Review**	**Test & Gate**	**Deploy**	**Monitor**	**Document**	**Learn**
Product / PM	AI + Human	AI (/avengers)	AI + parallel	AI + Human	AI-automated	Human initiates	NR MCP (AI)	AI auto-update	Golden patterns

↺ *repeats every sprint — knowledge compounds*

What AI handles vs. human:

- Human-led
- AI-led (Claude Code)
- NR MCP observability
- Auto-documentation

AI writes code, runs tests, reviews security, updates Jira — autonomously, in parallel.

Humans approve at 5 gates:
Requirements · Architecture · Security · PR · Deploy.

Each session makes the next one smarter.
Knowledge never leaves the team.

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Onboard in Hours, Not Weeks — Full Context on Day One

BEFORE

2 WEEKS

Slack DMs · git archaeology · tribal knowledge
→ still not fully productive by end of week 2

→

AFTER

2 HOURS

/bootstrap: detects stack, seeds memory,
builds code graph, wires MCPs — in 15 seconds

Full codebase context

Auto-loaded at session start.
No manual exploration needed.

Architecture diagram

Built from code graph.
Updates automatically.

Team's lessons ready

patterns + anti-patterns +
wins — from day 1.

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/avengers — Multi-Agent Orchestration · 5 Engineers in One Session

Complex tasks that used to take one engineer one full day:

Nick Fury (Opus — Orchestrator) *routes work · validates gates · blocks bad code*

Planner	Coder ×N	Reviewer	Validator
codebase research → implementation plan	parallel code writing verifies build passes	code standards + OWASP security	tests + blast radius before any merge

1. Fury reads spec → spawns Planner + Coders simultaneously
2. Each coder works on an independent file — zero waiting
3. Reviewer checks all output → Validator runs tests + blast radius
4. Engineer reviews 1 PR instead of writing 5 files

1 DAY

→

2 HOURS

Multi-file Airflow DAG + dbt model + NR alert working in parallel

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Incident Investigation — AI Queries New Relic Directly

MANUAL

2–3

HOURS

Log in to NR UI · navigate entity · grep logs
switch tabs · compare time ranges · Slack team

Form hypothesis → test → repeat → still not sure

Meanwhile: customer impact growing every minute

→

NR MCP

30

MINS

Claude queries NR via MCP · structured analysis
consumer lag, throughput, traces — all in one call

"analyze kafka consumer lag last 4h" → root cause

12 NR MCP tools available. AI knows what to call and when.

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Ship Without Surprises — Impact Analysis Before You Write a Line

We know what breaks BEFORE touching it. Not after deploying.

⚠ Before

Engineer changes spark_ingest.py.
Deploys to staging.
Downstream Kafka consumer breaks silently.
5-day incident to trace back the cause.

→

✅ After

Before writing code: get_impact_radius().
Sees 3 callers, 1 downstream Kafka consumer.
Fixes scope — ships confidently.
Zero prod incident.

~1 sprint/qtr

absorbed in unexpected rework

< 1 day

rework after impact-aware development

28 tools

in code-review-graph MCP — all graph-aware

Auto-updates

graph rebuilds on every file save

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Documentation That Writes Itself — Jira & Confluence Auto-Updated

Work session completes → /wrap-up fires → Claude drafts summary → Jira ticket updated via MCP → Confluence page published

Jira — What gets auto-updated

Status	In Review → Done — automatically
Comment	What was done, decisions made
Worklog	Time logged from session duration
Links	Commits linked via branch name

Confluence — What gets published

Architecture	Decisions logged after every session
Postmortems	Incident page auto-created from wrap-up
Release Notes	From git history — v2.2 → v2.3 auto
Runbooks	Updated as team learns new patterns

Zero manual effort. Every session documented. Every time.

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Knowledge That Compounds

Every session makes every future session smarter

ALWAYS LOADED

hot.md (~2k tokens at session start)

Active priorities + recent lessons + architecture summary.
/wrap-up regenerates after every session.

PULL ON DEMAND

lessons.md · architecture.md · history.md

Team knowledge retrieved only when relevant — not loaded upfront.
get_memory() · search_memory() · list_lessons()

REPLAYABLE WINS

/golden save → /replay → 50× on repeat problems

Validated fixes distilled into reusable templates.
/replay loads proven steps — team stops re-deriving the same root cause.

Month 1: 8 issues/task → Month 3: 1–2 issues/task · The system gets sharper, not staler.

The Results — Numbers the Team Has Measured

10×

faster on novel feature work
with graph + memory context

50×

faster on repeat problems
using /golden + /replay

6×

reduction in MTTR
3 hours → 30 minutes

90%

test coverage achieved
vs 60–70% before

Additional measurements

2 weeks → 2 hours	14–20 days → 3–5 days	<1 day	0 manual updates
engineer onboarding	feature cycle time	rework from hidden deps	Jira + Confluence

Sustained team average: 15–25× productivity improvement · over one month of real data engineering work

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Before & After — Ingestion Team · 3 Repos · ~3 Months

What We Measured	Before	After (AI-Led)
Engineer onboarding	2+ weeks	**2 hours — context loads on day 1**
Incident MTTR	2–3 hours	**20–30 minutes**
Feature cycle time	14–20 days	**3–5 days**
Blast radius visibility	Unknown until prod breaks	**Instant — before coding starts**
Rework from hidden deps	~1 sprint / quarter	**< 1 day**
Security bugs caught	~40%	**80–100% (every PR automated)**
Test coverage	60–70%	**90% (gate enforced)**
Jira / Docs update time	Manual, often skipped	**Zero — auto-updated on /wrap-up**

Repos: om-airflow-dags · spark-kafka-apps · tf-dataos-new-relic

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AI Does the Work. Humans Stay in Control. Always.

1	**Requirements**	Human writes the spec. AI reads it.
2	**Architecture**	Human approves the plan before any code is written.
3	**Security Review**	AI flags every OWASP issue. Any critical finding blocks.
4	**PR Merge**	Human reviews every PR — agent output treated like human code.
5	**Deploy**	Human initiates. Human monitors. AI writes release notes.

Agent PRs are treated identically to human PRs — same review process, same standards, same gates.

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Getting Started — From Zero to Productive in 4 Steps

15 MIN

Step 1: Install

./install.sh + ./verify.sh

One-time setup per engineer. Prereqs auto-checked.

15 SEC

Step 2: /bootstrap

First visit to any repo

Detects stack, seeds memory, builds code graph, wires MCPs.

ONGOING

Step 3: Work

Skills auto-trigger as you type

20 skills activate on file type or topic. No manual setup.

30 SEC

Step 4: /wrap-up

Every session end

Persists history, refreshes graph, updates Jira, rebuilds hot.md.

Week 1

Install + bootstrap 1 repo + /wrap-up every session

Week 2

All active repos. First /golden save after a win.

Week 3

/avengers on a real multi-file task. Golden library: 3–5 patterns.

Week 4+

Team shares goldens. Lessons compound. 15–25× productivity.

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What We're Asking

Support adoption	Drive AI-led workflows across all Data Engineering in DTM
Allocate time	Engineers need 30 min/day: /wrap-up + goldens. It compounds.
Endorse the model	Establish this as the template for all DE teams to adopt

15–25×
sustained avg

10×
novel work

50×
repeat tasks

30 min
to root cause

Thank You · Questions?

Repo: github.com/nr-mdakilahmed/claude_code_setup